Practice Set A

# Class 12 Mathematics, Chapter 8: Application of Integrals

Maximum marks: **53** **20** questions

Name _____ Date _____ Score _____ / 53

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1

8 marks

**1** MCQ, CBSE SQP 2024-25 — 1 mark

A student observes an open-air honeybee nest on the branch of a tree, whose plane figure is parabolic in shape, given by $x^2 = 4y$. Then the area (in sq units) of the region bounded by the parabola $x^2 = 4y$ and the line $y = 4$ is:

**(A)** $\frac{32}{3}$

**(B)** $\frac{64}{3}$

**(C)** $\frac{128}{3}$

**(D)** $\frac{256}{3}$

**2** MCQ, Board-pattern — 1 mark

The area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = 36$ and the coordinate axes is:

**(A)** $9\pi$ sq units **(B)** $18\pi$ sq units

**(C)** $36\pi$ sq units **(D)** $6\pi$ sq units

**3** MCQ, Board-pattern — 1 mark

The area enclosed by the ellipse $x^2 + 4y^2 = 16$ is:

**(A)** $16\pi$ sq units **(B)** $64\pi$ sq units

**(C)** $8\pi$ sq units **(D)** $4\pi$ sq units

**4** MCQ, Board-pattern — 1 mark

The area of the region bounded by the curve $y = \sin x$, the x-axis and the lines $x = 0$ and $x = \pi$ is:

**(A)** $1$ sq unit **(B)** $2$ sq units

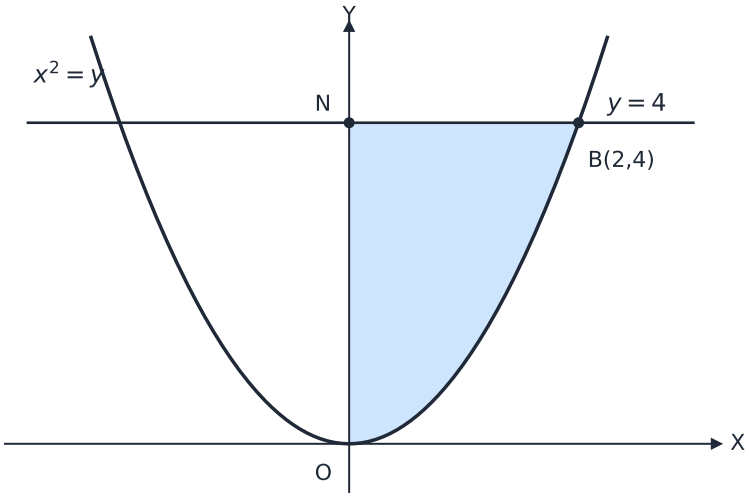
**(C)** $0$ **(D)** $\pi$ sq units

**5** MCQ, Board-pattern — 1 mark

The area of the region bounded by the line $x + 2y = 4$ and the coordinate axes is:

**(A)** $8$ sq units **(B)** $2$ sq units

**(C)** $16$ sq units **(D)** $4$ sq units

| Q No. | Question | Marks |
|---|---|---|
| 6 | Assertion-Reason, Board-pattern<br>**Assertion (A):** The area of the region bounded by the curve $y = x^3$, the x-axis and the lines $x = -1$ and $x = 1$ is $0$. **Reason (R):** $\int_{-1}^{1} x^3\,dx = 0$, as $x^3$ is an odd function.<br>**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).<br>**(C)** Assertion (A) is true, but Reason (R) is false.<br>**(D)** Assertion (A) is false, but Reason (R) is true. | 1 mark |
| 7 | Short answer, CBSE SQP 2025-26<br>Find the area of the shaded region in the figure given below. The region lies in the first quadrant and is bounded by the parabola $x^2 = y$, the y-axis and the line $y = 4$.<br>  | 2 marks |

## Level 2

18 marks

| | | |
|---|---|---|
| 8 | Short answer, PYQ-style<br>Sketch the graph of $y = x^2$ and, using integration, find the area of the region bounded by the curve $y = x^2$, the x-axis and the lines $x = -1$ and $x = 2$. | 3 marks |
| 9 | Short answer, PYQ-style<br>Sketch the region bounded by the lines $2x + y = 8$, $y = 2$, $y = 4$ and the y-axis. Hence, obtain its area using integration. | 3 marks |
| 10 | Short answer, PYQ-style<br>Using integration, find the area of the region bounded by the line $y = 4 - 2x$, the x-axis and the ordinates $x = -1$ and $x = 3$. | 3 marks |
| 11 | Short answer, Board-pattern<br>Using integration, find the area of the region bounded by the curve $y = \frac{3}{2}\sqrt{4 - x^2}$ and the x-axis. | 2 marks |

| Q No. | Question | Marks |
|---|---|---|
| 12 | Short answer, PYQ-style<br>Using integration, find the area of the minor segment of the circle $x^2 + y^2 = 16$ cut off by the line $x = 2$. | 3 marks |
| 13 | Short answer, CBSE SQP 2024-25<br>Draw the rough sketch of the curve $y = 20\cos 2x$, where $\frac{\pi}{6} \le x \le \frac{\pi}{3}$. Using integration, find the area of the region bounded by the curve $y = 20\cos 2x$ from the ordinates $x = \frac{\pi}{6}$ to $x = \frac{\pi}{3}$ and the x-axis. | 2 marks |
| 14 | Short answer, PYQ-style<br>Using integration, find the area of the region bounded by the curve $y = x\|x\|$, the x-axis and the ordinates $x = -1$ and $x = 1$. | 2 marks |

## Level 3 — 27 marks

| | | |
|---|---|---|
| 15 | Long answer, CBSE SQP 2026-27<br>Using integration, find the area of the region $\{(x, y) : 0 \le y \le \sqrt{4 - x^2},\ x \ge 2 - y\}$. | 5 marks |
| 16 | Long answer, PYQ-style<br>Using integration, find the area of the triangle whose vertices are $A(-1, 0)$, $B(1, 3)$ and $C(3, 2)$. | 5 marks |
| 17 | Long answer, CBSE SQP 2026-27<br>Using integration, find the area of the region bounded by the curve $y^2 = 2x$ and the line $x - 2y = 0$. | 5 marks |
| 18 | Long answer, CBSE SQP 2026-27<br>Using integration, find the area of the region enclosed within $25x^2 + 16y^2 = 400$. | 5 marks |
| 19 | Short answer, CBSE SQP 2025-26<br>Using integration, find the area of the region $\{(x, y) : x^2 - 4y \le 0,\ y - x \le 0\}$. | 3 marks |
| 20 | Case study, Board-pattern<br>**Case Study:** A circular garden in a housing society has its boundary given by $x^2 + y^2 = 100$, with its centre at the origin (units in metres). A flower bed is to be made in the part of the garden between two straight paths along the lines $x = -5$ and $x = 5$. Based on this information, answer the following questions using integration. | 4 marks |

**(i)** Find the total area of the garden. — 1 mark

**(ii)** Find the area of the part of the garden lying in the first quadrant. — 1 mark

**(iii)** Find the area of the flower bed, that is, the part of the garden between the lines $x = -5$ and $x = 5$. — 2 marks

# Answer key

Chapter 8: Application of Integrals, Practice Set A. Full step-wise solutions are in the solutions sheet.

## Level 1

| | |
|---|---|
| 1 | (B) $\frac{64}{3}$ |
| 2 | (A) $9\pi$ sq units |
| 3 | (C) $8\pi$ sq units |
| 4 | (B) $2$ sq units |
| 5 | (D) $4$ sq units |
| 6 | (D) A is false, but R is true |
| 7 | $\frac{16}{3}$ sq units |

## Level 2

| | |
|---|---|
| 8 | $3$ sq units |
| 9 | $5$ sq units |
| 10 | $10$ sq units |
| 11 | $3\pi$ sq units |
| 12 | $\left(\frac{16\pi}{3} - 4\sqrt{3}\right)$ sq units |
| 13 | $(20 - 10\sqrt{3})$ sq units |
| 14 | $\frac{2}{3}$ sq units |

## Level 3

| | |
|---|---|
| 15 | $(\pi - 2)$ sq units |
| 16 | $4$ sq units |
| 17 | $\frac{16}{3}$ sq units |
| 18 | $20\pi$ sq units |
| 19 | $\frac{8}{3}$ sq units |
| 20 | (i) $100\pi\ \mathrm{m}^2$ (ii) $25\pi\ \mathrm{m}^2$ (iii) $\left(50\sqrt{3} + \frac{100\pi}{3}\right)\ \mathrm{m}^2$ |